

AI-Assisted Citation Integrity Audit

Topic: Failure Modes in Chained Tool-Calling Agents Operating Without Human Checkpoints

Student Detail	Fill in
Name	Sachin Kumar
Roll Number	MRA2026004
Programme / Department	Information Technology (Robotics and AI)
Topic ID	T9
Full Assigned Topic	Failure Modes in Chained Tool-Calling Agents Operating Without Human Checkpoints
AI Tool Used	ChatGPT
Date	21 august, 2026

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Almost daily

A2. Have you previously asked an AI system to generate references?

Once or twice

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Rarely

A4. Before this lab, how confident are you that citations generated by AI are

generally reliable? ☐ 4

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Chatgpt
Model name	GPT-5.6 Luna

Model/version shown	GPT-5.6 Luna
Access tier	☐ Free

Web browsing available?	☐ Yes
Was web browsing actually used?	☐ Yes
Research/Deep Research mode used?	☐ No
Date/time generated	21 August, 2026

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: “[YOUR ASSIGNED TOPIC]”. The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental

evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title:

Measure	Value
Approximate pages	15
Approximate word count	5000
Number of sections	8
Total number of references	22
Approximate in-text citation occurrences	40
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☒ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	11
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	10
URL only	0
No persistent identifier supplied	1
TOTAL	22

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning	arXiv ID available

	and Acting in Language Models. arXiv:2210.03629	
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents

selection bias. If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	7
R5	11
R6	14
R7	17
R8	20
R9	21

R10	22
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Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: "If I had not verified this citation, how convincing would it look to me?"

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure

4 Looks genuine

5 Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Looks convincing and genuine, with correct details	5	Yes / No
R2	Not complete	4	Yes / No
R3	Correct details	5	Yes / No
R4	Looks genuine and realistic authors	5	Yes / No
R5	Looks genuine and realistic authors	5	Yes / No
R6	Looks genuine and realistic authors	4	Yes / No
R7	Looks genuine and realistic authors	5	Yes / No
R8	Looks genuine and realistic authors	5	Yes / No

R9 Looks genuine and realistic

authors⁵ Yes / No

R10 Looks genuine and realistic

authors⁵ Yes / No

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are

valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

- ☐ Persistent identifier: DOI / arXiv / PMID
- ☐ Publisher or official repository
- ☐ Google Scholar
- ☐ Semantic Scholar / OpenAlex
- ☐ Crossref
- ☐ PubMed or another discipline-specific database
- ☐ Exact-title search
- ☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	N	Y	Y	Y
R2	Y	Y	Y	Y/	Y/N	Y
R3	Y	Y	Y	Y	Y/N	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	N	Y	Y	Y
R8	Y	Y	Y	Y	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	https://arxiv.org/	arXiv:2410.09024	Publications exist with matching information.
R2	Springer.com	DOI: 10.1007/s10462-025-11471-9	Publications exist with matching information.
R3	https://arxiv.org/	arXiv:2406.13352.	Publications exist with matching information.
R4	https://arxiv.org/	arXiv:2308.03688	Publications exist with matching information.
R5	https://arxiv.org/	arXiv:2303.11366	Publications exist with matching information.
R6	springer.com	DOI 10.18653/v1/2025.findings-e mnlp.958	Publications exist with matching information.

R7 <https://arxiv.org/> arXiv:2410.02644 Publications exist with matching information.

R9 [springer.com](https://www.springer.com) DOI: 10.1145/3711896.3736570
Publications exist with matching information.

R8 <https://arxiv.org/>

arXiv:2305.04388; DOI
10.52202/075280-3275

Publications exist with matching information.

R10 [springer.com](https://www.springer.com) DOI: 10.18653/v1/2025 Publications exist with matching information.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	B	the AI omitted Eric Hambro from the author list.
R2	A	All details correct
R3	A	All details correct
R4	A	All details correct
R5	A	All details correct

R6	A	All details correct
R7	A	All details correct
R8	A	All details correct
R9	A	All details correct
R10	A	All details correct

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 9 B = 1 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1

D	0
E	1

Authenticity Points = $4A + 3B + C + E$

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(\underline{9}) + 3(\underline{1}) + 1(\underline{0}) + 0(\underline{0}) + 1(\underline{0}) = \underline{39}$ points

Authenticity Score = $\underline{39} / (4 \times \underline{10}) \times 100 = \underline{97.5} / 100$

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 10 Total predictions = 10 Prediction Accuracy = 100 %

FINAL LAB 1 RESULTS

Measure	Your Result
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Total references in AI paper	22
References deeply audited	10
A: Verified	9
B: Wrong metadata	1
C: Frankenstein	0
D: Fabricated	0

E: Identifier mismatch	0
Authenticity Score	_97.5___ /100
Prediction Accuracy	___97.5_ %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

.Yes most of these turn out to be genuine, but one citation has an error2. Did all genuine references actually support the claims for which they were cited?

Yes3. What was the most serious citation-integrity problem you observed?

.incomplete author detail4. What is the single most important lesson you learned from this lab?

A professional looking citations can be actually false

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: ..Sachin Kumar

Roll Number: MRA2026004.....

Signature: .Sachin.....

Date: .21 August 2026.....

CENTRAL LESSON OF LAB 1

The audit produced a high authenticity score of 97.5/100, with 9 fully verified references and 1 reference with wrong metadata.

The most important distinction is that 97.5/100 does not mean that every citation in the complete 22-reference bibliography has been verified. Only the required 10-reference sample was deeply audited. The remaining 12 references would require separate verification before the bibliography could be described as fully validated.